I Can find the rule in a table

Level 3
Algebra
Teacher 1

CCSS 3.OA.D.9 · **EE** M.EE.3.OA.8 · **CRA:** Concrete → Pictorial · **Flow:** Standard (2 new routine steps)

Lesson guidance:

Tip: Read across. Read a row across, In then Out. Never down the Out column.

Tip: Say the rule. Pupils write a sign and a number. They SAY "add 3" or "times 2".

Watch: One row is not enough. 2 → 4 fits + 2 and × 2. Test every row, aloud.

Lesson flow:

#	Section	Pupil page	Min
1	Prerequisite Check	Prerequisite Check	5
2	Number sense	Number sense	5
3	Spaced retrieval	Mini whiteboards	3
4	Chaining bridge	Mini whiteboards	1
5	I Do: the anchor chart	Anchor chart	10
6	We Do: Guided Practice	Lesson sheet	8
7	Check Question	Mini whiteboards	3
8	You Do: Independent Practice	Practice 1–2	12
9	Word problem	Word problem	5
10	Plenary and exit slip	Exit slip	4
	Transitions	between pages	4
	Total		**60**

Materials:

- ☐ A paper "rule machine" box with In and Out slots
- ☐ Reading strips (card to cover the other rows)
- ☐ Pupil packet, one per pupil
- ☐ Number cards 1 to 20
- ☐ Mini whiteboards and pens
- ☐ Exit slips, cut in half

Vocabulary:

4 →

In

the number that goes in

→ 7

Out

the number that comes out

4 → + 3 → 7

rule

what the machine does to every number

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I Can find the rule in a table

Level 3
Algebra
Teacher 2

What students need to know:

#	Skill or Concept	Atom Type	Teacher Notes
1	Add facts to 20	Fact · Pre-requisite	Find a change fast. Prerequisite Check a–d.
2	Subtract facts to 20	Fact · Pre-requisite	Prerequisite Check e–h.
3	×2, ×5, ×10 facts	Fact · Pre-requisite	Prerequisite Check i–l.
4	**In and Out:** each row is a pair	Category	Read across, not down.
5	Bigger or smaller	Comparative	Out bigger: + or ×. Out smaller: −.
6	**Test every row**	**Conceptual**	**Critical Atom. A rule from one row may be wrong (2 → 4).**
7	+ or ×	Comparative	+ fails on row 2? Try ×.

What students do:

Step	Task / Routine atom	Tag	Check it with
1	Read In and Out across row 1.	New	"Point to In. Point to Out."
2	Find the change: bigger or smaller, by how much.	Pre-requisite	Prerequisite Check {pre.0.letters}, {pre.1.letters}
3	**Test the rule on every row.**	**New**	**Check Question; exit slip c**
4	Write the sign in the circle, the number in the box.	Pre-requisite	Guided cell 2

Rows tagged New: 2 → Flow: Standard. One chunk: + rules, then − rules, then × rules; test every row.

How to teach:

Feed cards through the rule machine first. Model + (the change is plain), then −, then a × table where "+ the change" fails on row 2: the failure is the lesson. Say the test aloud on every row.

Board: The In / Out table 2 → 4, 9 → 18, 10 → 20. Write "Wrong" beside + 2.

Common mistakes:

Mistake	Step	Looks like	Fix	Caught by
One pair only	3	+ 4 for 1 → 5, 3 → 15, 4 → 20	Test row 2 aloud.	Check Question A
Down the Out column	1	+ 10 for 1 → 5, 3 → 15, 4 → 20	Reading strip: one row across.	Check Question B
Sign flipped	2	+ 4 for 6 → 2, 8 → 4, 13 → 9	"Bigger or smaller?"	Exit b · Practice 1

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I Can find the rule in a table

Level 3
Algebra
Reference

Reference key:

Pages and order

Page	Tab	Score
P0	Prerequisite	one per question
P1	Number sense	/10
P2	Anchor chart	none (a poster)
P3	Lesson 1	Independent cells
P4	Practice n	every cell
P5	Word problem	/1
P6	Exit slip	/3 a half
T1–T5	Teacher n	—
T6	Answer Key	—

Fade levels

Label	Means
★	Guided: hints on, step 1 traced
★★	Independent: hints off
★★★	Last row: the edge cases

Data loop

● **Most:**	Go on.
◑ **About half:**	Short reteach, then go on.
○ **Fewer than half:**	Stop. Reteach with a concrete example.

Abbreviations

Short	In full
MWBs	mini whiteboards
ELL	English language learners
SPED	special education students
RCDWS	Read, Circle, Draw, Write, Solve
CRA	Concrete-Representational-Abstract

Routing ids

No site lesson exists yet for these prerequisites: T6 names a CCSS code and a short title (V22 Appendix H).

Answer slots

Shape	Used for
_____	a number after =
☐	a part inside an expression
☐	a tick (routine panel, exit slip)
rule frame	the sign in the circle, the number in the box

Icons

Icon	Means
	Tip
	Feedback
	Watch
	Research
	Critical Atom
	Language tip
	Why it works
	Board
1 2 3 4	The pupil Steps (accent)

Band labels

Pupil: Vocabulary · Remember · Guided Practice · Steps · Independent Practice · Mixed Review · Number sense: A / B · Workspace · Another example · Say · Rule.

Teacher: Lesson guidance · Lesson flow · Materials · What students need to know · What students do · How to teach · Common mistakes · Section n · Prerequisite routing · Accepted answers · Answers at a glance.

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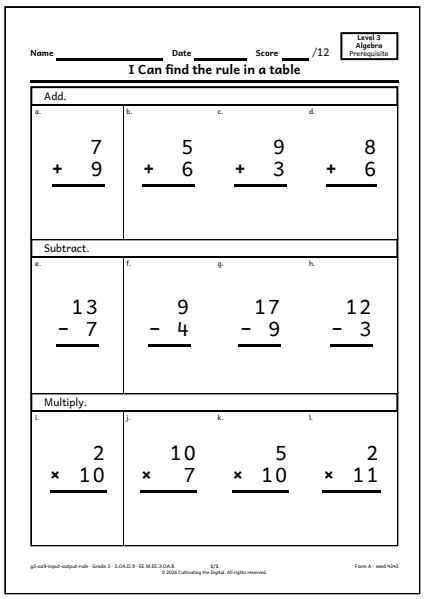
I Can find the rule in a table

Level 3
Algebra
Teacher 3

Section 1 · Prerequisite Check (5 min)

> **Tip: Route before teaching.** Mark it while pupils do the Number sense page.

1. **Do:** Hand out the Prerequisite Check, face down.
2. **Say:** "Do these on your own. Try every one."
3. **Do:** Walk and mark as pupils finish.
   Note: Expected: **a. 16 · b. 11 · c. 12 · d. 14 · e. 6 · f. 5 · g. 8 · h. 9 · i. 20 · j. 70 · k. 50 · l. 22**. Routing is on the last page (T6).



Name _____ Date _____ Score ___ /12 — Level 3 Algebra Prerequisite

I Can find the rule in a table

Add.

a.	b.	c.	d.
7 + 9	5 + 6	9 + 3	8 + 6

Subtract.

e.	f.	g.	h.
13 − 7	9 − 4	17 − 9	12 − 3

Multiply.

i.	j.	k.	l.
2 × 10	10 × 7	5 × 10	2 × 11

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Section 2 · Number sense (5 min)

1. **Do:** Hand out the Number sense page.
2. **Say:** "Band A: read each row. Write the answer."
3. **Ask:** "What is 1 less than 40?"
   Note: Expected: **39**. Band B: the same five kinds, harder numbers.
4. **Ask:** "What is 10 more than 305?"
   Note: Expected: **315**.

Section 3 · Spaced retrieval (3 min)

1. **Do:** Pupils hold mini whiteboards. Write each question on the board.
2. **Ask:** "10 × 6 = ?"
   Note: Expected: **60**.
3. **Ask:** "18 − 9 = ?"
   Note: Expected: **9**.
4. **Ask:** "What is 10 more than 45?"
   Note: Expected: **55**.

Section 4 · Chaining bridge (1 min)

1. **Say:** "Yesterday each times row went up by the same number."
2. **Say:** "Today the machine has a rule. We find it."

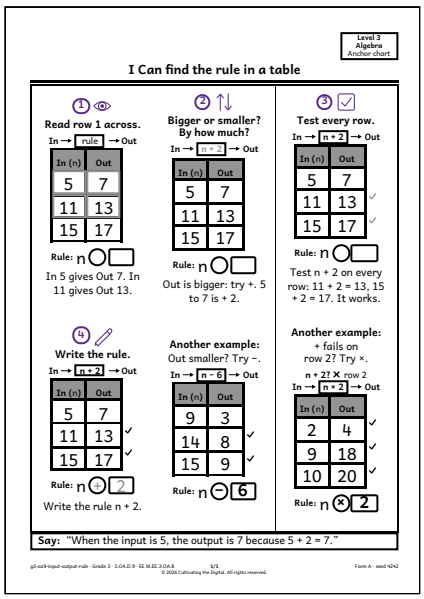
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I Can find the rule in a table

Level 3
Algebra
Teacher 4

Section 5 · I Do: the anchor chart (10 min)

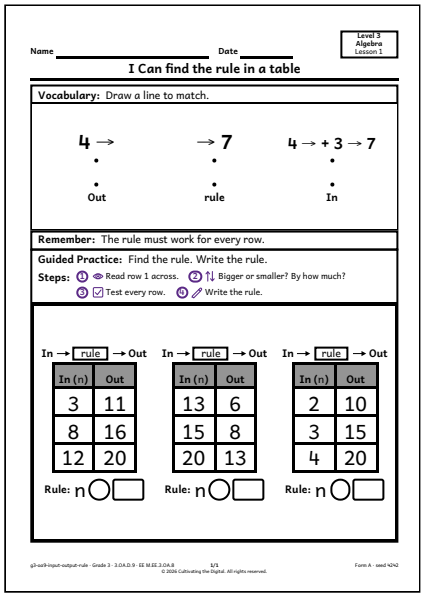
Watch: The nodders. Pupils who nod at + 2 on panel 3 are the one-pair group.



1	**Do:**	Feed card 5 into the rule machine. Show 7.
2	**Say:**	"In 5, out 7. It got bigger."
	Note:	Step 1 and step 2.
3	**Ask:**	"Bigger by how much?"
	Note:	Expected: **2**.
4	**Say:**	"Test every row: 11 + 2 is 13."
	Note:	Step 3. Then row 3. The rule: **+ 2**.
5	**Say:**	"In 9, out 3. It got smaller."
	Note:	Panel 2. The rule: **− 6**.
6	**Say:**	"In 2, out 4. Plus 2? Test row 2."
	Note:	Panel 3: the trap.
7	**Say:**	"9 plus 2 is 11. Not 18. Try times."
	Note:	Write "Wrong" beside + 2 on the board only.
8	**Say:**	"The rule is times 2. It works for every row."
	Note:	Pupils repeat. The rule: **× 2**.

Section 6 · We Do: Guided Practice (8 min)

Feedback: Fix it at once. Fix: cover the other rows with the strip. "Test this row."



1	**Say:**	"Look at 3 → 11, 8 → 16, 12 → 20. Read row 1 across with me."
2	**Ask:**	"Bigger or smaller? By how much?"
	Note:	Expected ★: **+ 8**.
3	**Do:**	Pairs do 13 → 6, 15 → 8, 20 → 13: A reads a row, B tests it.
	Note:	Expected ★: **− 7**.
4	**Ask:**	"Cell 2: is Out bigger or smaller?"
	Note:	Smaller, so −. Expected: **− 7**.
5	**Say:**	"In ____, out ____. The rule is ____."
	Note:	The Say frame, for every row.
6	**Do:**	Cell 3, 2 → 10, 3 → 15, 4 → 20: test + 8 on row 2 together.
	Note:	It fails. Expected ★: **× 5**.

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I Can find the rule in a table

Level 3
Algebra
Teacher 5

Section 7 · Check Question (3 min)

1. **Do:** Write the table on the board: 1 → 5, 3 → 15, 4 → 20. A: + 4 · B: + 10 · C: × 5.

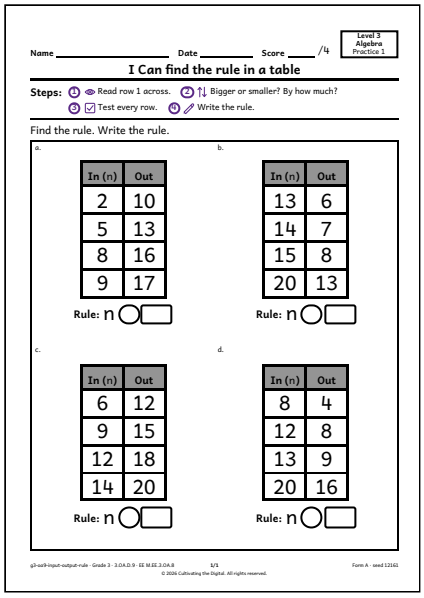
2. **Ask:** "What is the rule? Show fingers."
   Note: Correct: **C × 5**. Each wrong option is one T2 mistake: A: One pair only · B: Down the Out column

- ● **Most:** Go on to You Do.
- ◑ **About half:** Reteach "test every row" with the machine, then two more.
- ○ **Fewer than half:** Stop. Feed 9 through the machine. Test + 2 together.

Section 8 · You Do: Independent Practice (12 min)

Watch: The last rows. Walk to the last rows first: Practice 1: n + 6 (row 1 also fits × 2), n − 4 (row 1 also fits ÷ 2); Practice 2: n × 5 (row 1 also fits +), n − 20 (row 1 also fits ÷ 2).

1. **Do:** Pupils finish the lesson sheet, then Practice 1 and 2.
2. **Say:** "The Steps are at the top of the page."
3. **Ask:** "Does it work on row 3?"
   Note: Visit the one-pair group first.
4. **Do:** Practice 2: In numbers out of order. Read each row across.
   Note: Look for the gap between Outs written as the rule.



Name _____ Date _____ Score ___/4 Level 3 Algebra Practice 1

I Can find the rule in a table

Steps: 1 Read row 1 across. 2 Bigger or smaller? By how much? 3 Test every row. 4 Write the rule.

Find the rule. Write the rule.

a.

In (n)	Out
2	10
5	13
8	16
9	17

Rule: n ○ □

b.

In (n)	Out
13	6
14	7
15	8
20	13

Rule: n ○ □

c.

In (n)	Out
6	12
9	15
12	18
14	20

Rule: n ○ □

d.

In (n)	Out
8	4
12	8
13	9
20	16

Rule: n ○ □

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Section 9 · Word problem (5 min)

1. **Do:** Read the story aloud two times. Pupils tick each box.
2. **Ask:** "Boxes in, cars out. What is the rule?"
   Note: Expected: **× 2**, tested on every row.
3. **Ask:** "How many cars for 16 boxes?"
   Note: Expected: **32** cars.

Section 10 · Plenary and exit slip (4 min)

1. **Say:** "The rule must work for every row."
   Note: Say it together twice.
2. **Do:** Hand out the exit slip. No help. Sort the slips by item c.
   Note: a. **+ 8** · b. **− 4** · c. **× 5**.

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Name _____ Date _____ Score _____ /12

Level 3
Algebra
Prerequisite

I Can find the rule in a table

Add.

a.

$$\begin{array}{r} 7 \\ +\ 9 \\ \hline \end{array}$$

b.

$$\begin{array}{r} 5 \\ +\ 6 \\ \hline \end{array}$$

c.

$$\begin{array}{r} 9 \\ +\ 3 \\ \hline \end{array}$$

d.

$$\begin{array}{r} 8 \\ +\ 6 \\ \hline \end{array}$$

Subtract.

e.

$$\begin{array}{r} 13 \\ -\ 7 \\ \hline \end{array}$$

f.

$$\begin{array}{r} 9 \\ -\ 4 \\ \hline \end{array}$$

g.

$$\begin{array}{r} 17 \\ -\ 9 \\ \hline \end{array}$$

h.

$$\begin{array}{r} 12 \\ -\ 3 \\ \hline \end{array}$$

Multiply.

i.

$$\begin{array}{r} 2 \\ \times\ 10 \\ \hline \end{array}$$

j.

$$\begin{array}{r} 10 \\ \times\ 7 \\ \hline \end{array}$$

k.

$$\begin{array}{r} 5 \\ \times\ 10 \\ \hline \end{array}$$

l.

$$\begin{array}{r} 2 \\ \times\ 11 \\ \hline \end{array}$$

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Name _____ Date _____ Score _____ /10

Level 3
Algebra
Number sense

I Can find the rule in a table

Number sense: A

1 Write the missing number.

1 less than **40** is _____ .

2 Write the missing number.

10 more than **60** is _____ .

3 Write <, > or = in the circle.

72 ◯ **35**

4 Write how many of each place.

39 = ☐ tens ☐ ones

5 Write the missing number.

10 less than **44** is _____ .

Number sense: B

6 Write the missing number.

1 less than **117** is _____ .

7 Write the missing number.

10 more than **305** is _____ .

8 Write <, > or = in the circle.

544 ◯ **565**

9 Write how many of each place.

84 = ☐ tens ☐ ones

10 Write the missing number.

10 less than **572** is _____ .

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Level 3
Algebra
Anchor chart

I Can find the rule in a table

1 Read row 1 across.

In → rule → Out

In (n)	Out
5	7
11	13
15	17

Rule: n ◯ ▢

In 5 gives Out 7. In 11 gives Out 13.

2 Bigger or smaller? By how much?

In → n + 2 → Out

In (n)	Out
5	7
11	13
15	17

Rule: n ◯ ▢

Out is bigger: try +. 5 to 7 is + 2.

3 Test every row.

In → n + 2 → Out

In (n)	Out	
5	7	
11	13	✓
15	17	✓

Rule: n ◯ ▢

Test n + 2 on every row: 11 + 2 = 13, 15 + 2 = 17. It works.

4 Write the rule.

In → n + 2 → Out

In (n)	Out	
5	7	
11	13	✓
15	17	✓

Rule: n + 2

Write the rule n + 2.

Another example:

Out smaller? Try −.

In → n − 6 → Out

In (n)	Out	
9	3	
14	8	✓
15	9	✓

Rule: n − 6

Another example:

+ fails on row 2? Try ×.

n + 2? ✕ row 2

In → n × 2 → Out

In (n)	Out	
2	4	✓
9	18	✓
10	20	✓

Rule: n × 2

Say: "When the input is 5, the output is 7 because 5 + 2 = 7."

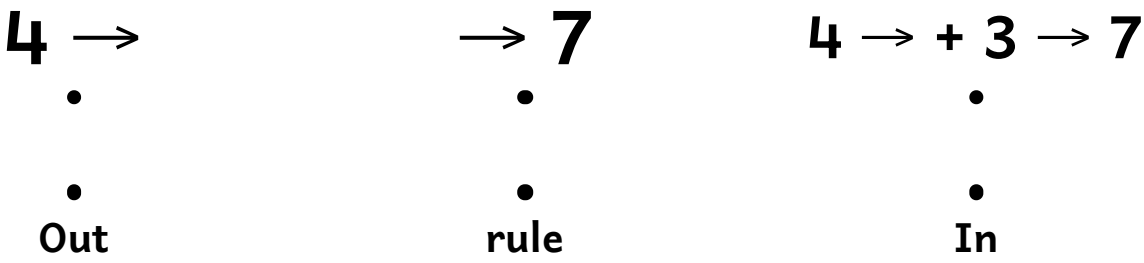
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Name _____ Date _____

Level 3
Algebra
Lesson 1

I Can find the rule in a table

Vocabulary: Draw a line to match.



Remember: The rule must work for every row.

Guided Practice: Find the rule. Write the rule.

Steps: (1) Read row 1 across. (2) Bigger or smaller? By how much? (3) Test every row. (4) Write the rule.

In → rule → Out

In (n)	Out
3	11
8	16
12	20

Rule: n ◯ ☐

In → rule → Out

In (n)	Out
13	6
15	8
20	13

Rule: n ◯ ☐

In → rule → Out

In (n)	Out
2	10
3	15
4	20

Rule: n ◯ ☐

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Name _____ Date _____ Score _____ /4

Level 3
Algebra
Practice 1

I Can find the rule in a table

Steps: (1) Read row 1 across. (2) Bigger or smaller? By how much?
(3) Test every row. (4) Write the rule.

Find the rule. Write the rule.

a.

In (n)	Out
2	10
5	13
8	16
9	17

Rule: n ◯ ☐

b.

In (n)	Out
13	6
14	7
15	8
20	13

Rule: n ◯ ☐

c.

In (n)	Out
6	12
9	15
12	18
14	20

Rule: n ◯ ☐

d.

In (n)	Out
8	4
12	8
13	9
20	16

Rule: n ◯ ☐

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Name _____ Date _____ Score _____ /4

Level 3
Algebra
Practice 2

I Can find the rule in a table

Steps: (1) Read row 1 across. (2) Bigger or smaller? By how much?
(3) Test every row. (4) Write the rule.

Find the rule. Write the rule.

a.

In (n)	Out
2	20
4	40
1	10
3	30

Rule: n ◯ ☐

b.

In (n)	Out
8	11
11	14
2	5
22	25

Rule: n ◯ ☐

c.

In (n)	Out
5	25
9	45
1	5
7	35

Rule: n ◯ ☐

d.

In (n)	Out
40	20
46	26
35	15
41	21

Rule: n ◯ ☐

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Name _____ Date _____ Score _____ /1

Level 3
Algebra
Word problem

I Can find the rule in a table

Read the story. Draw it. Solve.

A machine: 5 boxes make 10 cars.

8 boxes make 16 cars.

15 boxes make 30 cars.

How many cars do 16 boxes make?

16 boxes make _____ cars.

- ☐ Read it two times.
- ☐ Circle the numbers.
- ☐ Underline the question.
- ☐ Draw it.
- ☐ Write the equation.
- ☐ Solve. Check.

Workspace:



Say: "The rule is _____."

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Name _____ Score _____ /3

Level 3
Algebra
Exit slip

I Can find the rule in *a* table

Find the rule. Write the rule.

a.

In (n)	Out
2	10
3	11
7	15

The rule is:

n ◯ ☐

b.

In (n)	Out
6	2
8	4
13	9

The rule is:

n ◯ ☐

c.

In (n)	Out
5	25
9	45
10	50

The rule is:

n ◯ ☐

Say: "In 6, out 11. The rule is _____."

☐ I can do this. ☐ I need one more day.

Name _____ Score _____ /3

Level 3
Algebra
Exit slip

I Can find the rule in *a* table

Find the rule. Write the rule.

a.

In (n)	Out
2	10
3	11
7	15

The rule is:

n ◯ ☐

b.

In (n)	Out
6	2
8	4
13	9

The rule is:

n ◯ ☐

c.

In (n)	Out
5	25
9	45
10	50

The rule is:

n ◯ ☐

Say: "In 6, out 11. The rule is _____."

☐ I can do this. ☐ I need one more day.

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Answer Key _____ **Date** _____ **Score** _____ /12

Level 3
Algebra
Answer Key

I Can find the rule in a table

Add.

a.	b.	c.	d.
7 + 9 = **16**	5 + 6 = **11**	9 + 3 = **12**	8 + 6 = **14**

Subtract.

e.	f.	g.	h.
13 − 7 = **6**	9 − 4 = **5**	17 − 9 = **8**	12 − 3 = **9**

Multiply.

i.	j.	k.	l.
2 × 10 = **20**	10 × 7 = **70**	5 × 10 = **50**	2 × 11 = **22**

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Answer Key _____ Date _____ Score _____ /10

Level 3
Algebra
Answer Key

I Can find the rule in a table

Number sense: A

1 Write the missing number.

1 less than **40** is **39** .

2 Write the missing number.

10 more than **60** is **70** .

3 Write <, > or = in the circle.

72 **>** **35**

4 Write how many of each place.

39 = **3** tens **9** ones

5 Write the missing number.

10 less than **44** is **34** .

Number sense: B

6 Write the missing number.

1 less than **117** is **116** .

7 Write the missing number.

10 more than **305** is **315** .

8 Write <, > or = in the circle.

544 **<** **565**

9 Write how many of each place.

84 = **8** tens **4** ones

10 Write the missing number.

10 less than **572** is **562** .

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Level 3
Algebra
Answer Key

I Can find the rule in a table

① Read row 1 across.

In → rule → Out

In (n)	Out
5	7
11	13
15	17

Rule: n ◯ ▢

In 5 gives Out 7. In 11 gives Out 13.

② Bigger or smaller? By how much?

In → n + 2 → Out

In (n)	Out
5	7
11	13
15	17

Rule: n ◯ ▢

Out is bigger: try +. 5 to 7 is + 2.

③ Test every row.

In → n + 2 → Out

In (n)	Out	
5	7	
11	13	✓
15	17	✓

Rule: n ◯ ▢

Test n + 2 on every row: 11 + 2 = 13, 15 + 2 = 17. It works.

④ Write the rule.

In → n + 2 → Out

In (n)	Out	
5	7	
11	13	✓
15	17	✓

Rule: n + 2

Write the rule n + 2.

Another example:

Out smaller? Try −.

In → n − 6 → Out

In (n)	Out	
9	3	
14	8	✓
15	9	✓

Rule: n − 6

Another example:

+ fails on row 2? Try ×.

n + 2? ✗ row 2

In → n × 2 → Out

In (n)	Out	
2	4	✓
9	18	✓
10	20	✓

Rule: n × 2

Say: "When the input is 5, the output is 7 because 5 + 2 = 7."

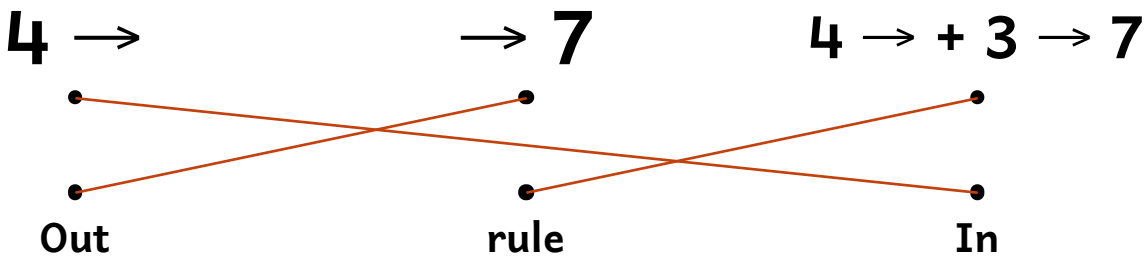
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Answer Key _____ Date _____

Level 3
Algebra
Answer Key

I Can find the rule in a table

Vocabulary: Draw a line to match.



Remember: The rule must work for every row.

Guided Practice: Find the rule. Write the rule.

Steps: (1) Read row 1 across. (2) Bigger or smaller? By how much? (3) Test every row. (4) Write the rule.

In → rule → Out

In (n)	Out
3	11
8	16
12	20

Rule: n + 8

In → rule → Out

In (n)	Out
13	6
15	8
20	13

Rule: n − 7

In → rule → Out

In (n)	Out
2	10
3	15
4	20

Rule: n × 5

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Answer Key _____ **Date** _____ **Score** _____ /4

Level 3
Algebra
Answer Key

I Can find the rule in a table

Steps: (1) Read row 1 across. (2) Bigger or smaller? By how much?
(3) Test every row. (4) Write the rule.

Find the rule. Write the rule.

a.

In (n)	Out
2	10
5	13
8	16
9	17

Rule: n + 8

b.

In (n)	Out
13	6
14	7
15	8
20	13

Rule: n − 7

c.

In (n)	Out
6	12
9	15
12	18
14	20

Rule: n + 6

d.

In (n)	Out
8	4
12	8
13	9
20	16

Rule: n − 4

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Answer Key _____ **Date** _____ **Score** _____ /4

Level 3
Algebra
Answer Key

I Can find the rule in a table

Steps: (1) Read row 1 across. (2) Bigger or smaller? By how much?
(3) Test every row. (4) Write the rule.

Find the rule. Write the rule.

a.

In (n)	Out
2	20
4	40
1	10
3	30

Rule: n × 10

b.

In (n)	Out
8	11
11	14
2	5
22	25

Rule: n + 3

c.

In (n)	Out
5	25
9	45
1	5
7	35

Rule: n × 5

d.

In (n)	Out
40	20
46	26
35	15
41	21

Rule: n − 20

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Answer Key _____ **Date** _____ **Score** _____ /1

Level 3
Algebra
Answer Key

I Can find the rule in a table

Read the story. Draw it. Solve.

A machine: 5 boxes make 10 cars.

8 boxes make 16 cars.

15 boxes make 30 cars.

How many cars do 16 boxes make?

16 boxes make **32** cars.

- ☐ Read it two times.
- ☐ Circle the numbers.
- ☐ Underline the question.
- ☐ Draw it.
- ☐ Write the equation.
- ☐ Solve. Check.

Workspace:

16 × **2** = **32**

Say: "The rule is **× 2** ."

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Answer Key _____ **Score** _____ /3

Level 3
Algebra
Exit slip

I Can find the rule in *a* table

Find the rule. Write the rule.

a.

In (n)	Out
2	10
3	11
7	15

The rule is: n + 8

b.

In (n)	Out
6	2
8	4
13	9

The rule is: n − 4

c.

In (n)	Out
5	25
9	45
10	50

The rule is: n × 5

Say: "In 6, out 11. The rule is + 5 ."

☐ I can do this. ☐ I need one more day.

Answer Key _____ **Score** _____ /3

Level 3
Algebra
Exit slip

I Can find the rule in *a* table

Find the rule. Write the rule.

a.

In (n)	Out
2	10
3	11
7	15

The rule is: n + 8

b.

In (n)	Out
6	2
8	4
13	9

The rule is: n − 4

c.

In (n)	Out
5	25
9	45
10	50

The rule is: n × 5

Say: "In 6, out 11. The rule is + 5 ."

☐ I can do this. ☐ I need one more day.

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I Can find the rule in a table

Level 3
Algebra
Answer Key

Prerequisite routing:

Question	Prerequisite (T2 atom)	If missed → Lesson
a–d. Add facts to 20	Fact, atom 1	2.OA.B.2 Add within 20
e–h. Subtract facts to 20	Fact, atom 2	2.OA.B.2 Subtract within 20
i–l. ×2, ×5, ×10 facts	Fact, atom 3	3.OA.C.7 Multiply by 2, 5 and 10
	Rule	0 missed: go on · 1: that lesson · 2+: the earliest

Accepted answers:

- The sign written + or "add"; × written x.
- One-step rules only: "+ 2 then …" is not the answer.
- Said aloud: "add 3", "plus 3" and "3 more" are all right.
- The number in the box with no sign in the circle: not yet right; ask "add or take away?".
- A right rule with no rows tested: right; ask the pupil to test it aloud.

Answers at a glance:

Page	Answers	Fade
P0 Prerequisite	**16 · 11 · 12 · 14 · 6 · 5 · 8 · 9 · 20 · 70 · 50 · 22**	★★
P1 Number sense	**39 · 70 · > · 3 tens 9 ones · 34 · 116 · 315 · < · 8 tens 4 ones · 562**	★★
P3 Guided	**+ 8 · − 7 · × 5**	★
P4-1 Practice, row 1	**+ 8 · − 7**	★★
P4-1 Practice, last row	**+ 6 · − 4**	★★★
P4-2 Practice, row 1	**× 10 · + 3**	★★
P4-2 Practice, last row	**× 5 · − 20**	★★★
P5 Word problem	**× 2 · 16 × 2 = 32**	★
P6 Exit slip	**a. + 8 · b. − 4 · c. × 5**	★★★

Wrong answers to look for when marking:

Pupil wrote	Likely mistake	Next step
+ 20 for 5 → 25, 9 → 45, 10 → 50	**One pair only**	Test row 2 aloud.
+ 1 for 2 → 10, 3 → 11, 7 → 15	**Down the Out column**	Reading strip: one row across.
+ 4 for 6 → 2, 8 → 4, 13 → 9	**Sign flipped**	"Bigger or smaller?" first.
+ 2 for 6 → 2, 8 → 4, 13 → 9	**Down the Out column**	Reading strip: one row across.
+ 5 for 3 → 11, 8 → 16, 12 → 20 (Guided 1)	**Down the Out column**	Cover the rows; read row 1 across.
+ 7 for 13 → 6, 15 → 8, 20 → 13 (Guided 2)	**Sign flipped**	Point: is Out smaller? Then −.
+ 8 for 2 → 10, 3 → 15, 4 → 20 (Guided 3)	**One pair only**	Test it on row 2 together.
+ 10 for 1 → 5, 3 → 15, 4 → 20 (Check Question)	**Down the Out column**	Cover the rows; read one across.
+ 5 for the story (Word problem)	**One pair only**	Test the rule on 8 boxes too.

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